

GitHub Projects Data Analytics Pipeline



Level 3 Project 2

Evaluation Rubric

Task 1: Data Collection & Preparation

Task Details:

- 1. Retrieve repository data from the GitHub API.
- 2. Convert the JSON response into a Pandas DataFrame.
- 3. Explore the dataset.
- 4. Select the required columns.
- 5. Prepare and clean the dataset.
- 6. Save and verify the cleaned dataset

Evaluation Criteria:

Task	Instructions	Passing Criteria
Collect API Data	Retrieve repository data from the GitHub API.	Repository data is retrieved successfully.
Create the Dataset	Convert the JSON response into a Pandas DataFrame and explore the dataset.	The dataset is loaded successfully and its structure is explored correctly.
Select Required Columns	Keep only the required columns for the analysis.	Only the required columns are retained.
Extract Required Values	Extract the required values from nested columns such as owner and license .	Required values are extracted correctly.
Handle Missing Values	Detect and handle missing values using an appropriate approach.	Missing values are handled correctly.
Handle Duplicate Records	Detect and remove duplicate records.	Duplicate records are removed successfully.
Prepare the Dataset	Convert date columns and rename columns to prepare the dataset for SQL analysis.	The dataset is formatted correctly for analysis.
Save & Verify the Dataset	Save the cleaned dataset as github_projects.csv and verify it.	The dataset is saved successfully and verified.

Task 2: Store & Analyse the Data

Task Details:

- 1. Store the cleaned dataset in SQLite.
- 2. Analyse the data using SQL.
- 3. Create Python visualizations.
- 4. Interpret the analysis results.

Evaluation Criteria:

Task	Instructions	Passing Criteria
Create the SQLite Database	Create the SQLite database and import the dataset into the Repositories table.	The database and table are created successfully.
Filtering & Searching Queries	Write SQL queries using filtering and searching techniques.	Queries execute successfully and return the correct results.
Logical Operators	Use AND , OR , and NOT where required.	Logical operators are applied correctly.
Sorting & Limiting Queries	Write SQL queries using sorting and limiting techniques.	Queries execute successfully and return the correct results.
Aggregate Functions	Apply aggregate functions such as COUNT and AVG .	Aggregate calculations are correct.
Grouping Analysis	Use GROUP BY and HAVING to summarize the data.	Grouping operations are performed correctly.
Create Python Visualizations	Create the required visualizations using Matplotlib.	Charts are created correctly and accurately represent the data.
Interpret Results	Explain the main findings from the SQL analysis and visualizations.	Findings are accurate and supported by the analysis.

Task 3: Version Control & Project Ethics

Task Details:

- 1. Manage the project using Git.
- 2. Upload the project to GitHub.
- 3. Complete the ethics reflection.

Evaluation Criteria:

Task	Instructions	Passing Criteria
Initialize & Check Repository	Initialize the Git repository and check the repository status.	Git commands are used correctly and the repository is initialized successfully.
Stage & Commit Files	Stage project files and create commits.	Git commands are used correctly and commits are created successfully.
Publish the Project on GitHub	Create a GitHub repository and push the project to GitHub.	The project is published successfully on GitHub.
Provide Project Evidence	Submit the required GitHub repository link and screenshots.	The required repository link and screenshots are provided successfully.
Complete the Ethics Reflection	Answer the required ethics reflection questions.	Responses demonstrate a clear understanding of responsible data practices.

